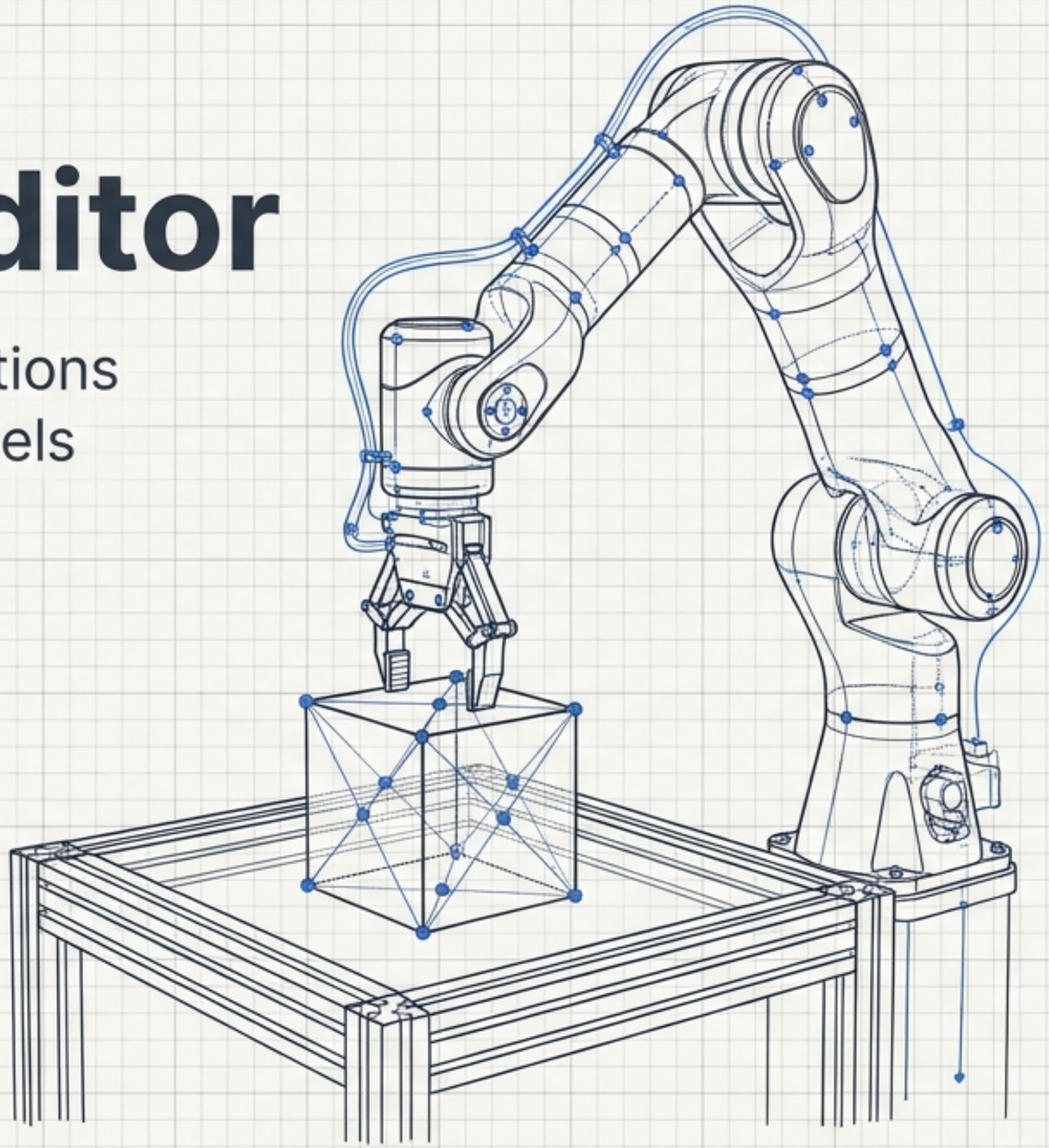
Physical Fidelity Auditor

Detecting Physics Hallucinations in 3D Generative World Models



Built for Nebius Physical AI Hackathon

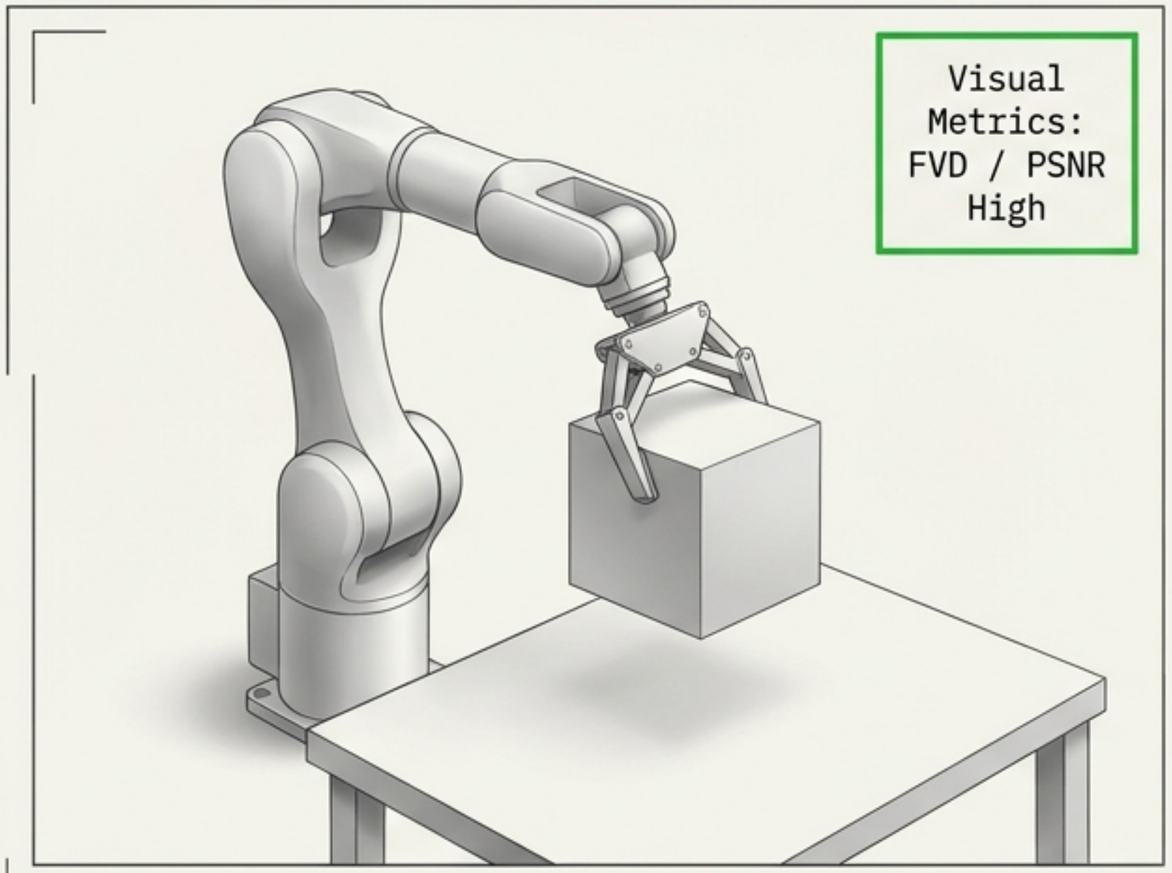
Open-Source MIT License

Gemini Notebook

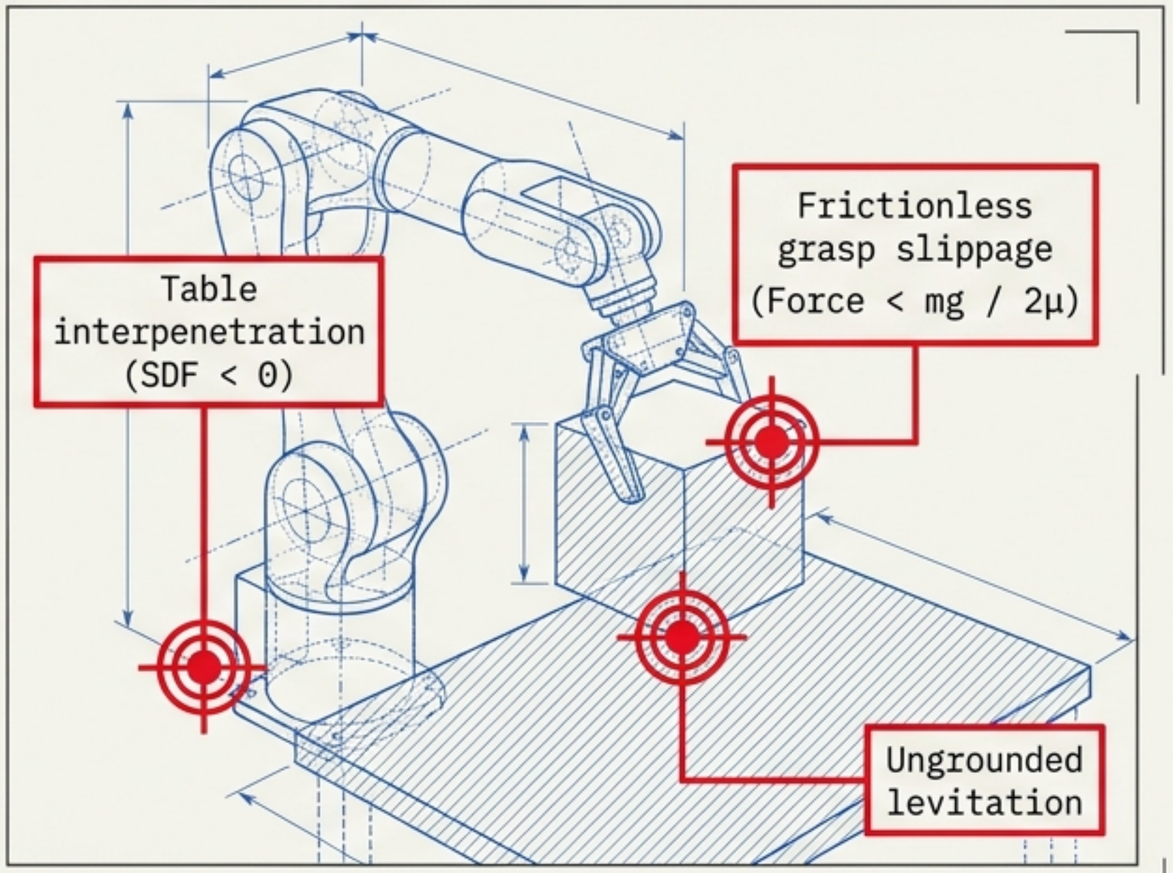
Generative world models optimize for pixels, leaving a critical gap in physical reality.

Current sim-to-real robotics lack measurement-based diagnostics. Models that appear visually solvent frequently hallucinate fundamental Newtonian mechanics.

The AI Dream



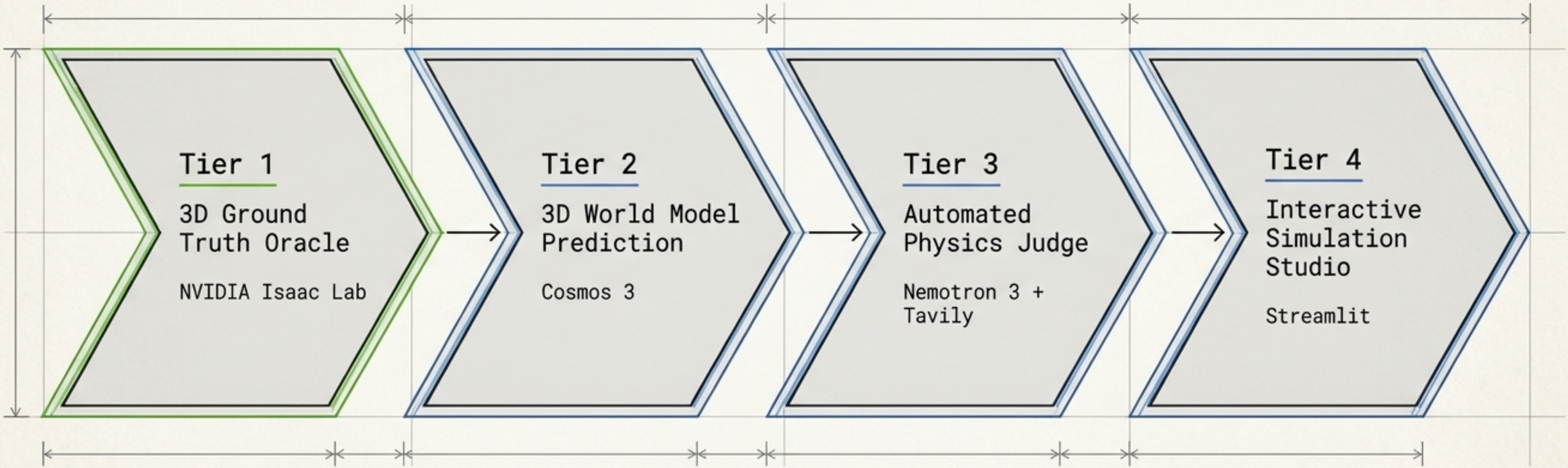
The Physical Reality



Gemini Notebook

The 4-Tier Physical Fidelity Auditor bridges the sim-to-real gap.

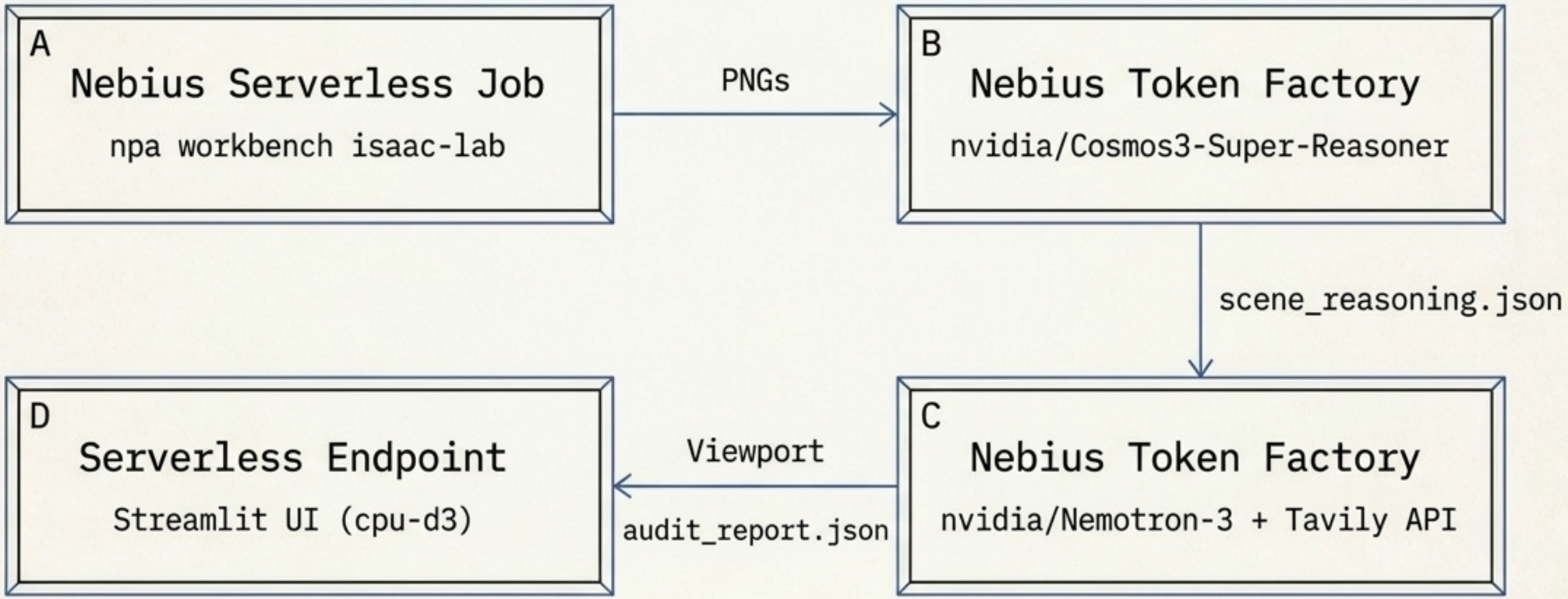
Deployed on Nebius AI Cloud, this automated diagnostic pipeline intercepts, evaluates, and grades generated robotic manipulations prior to physical deployment.



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A modular topology separating heavy simulation simulation from lightweight reasoning.

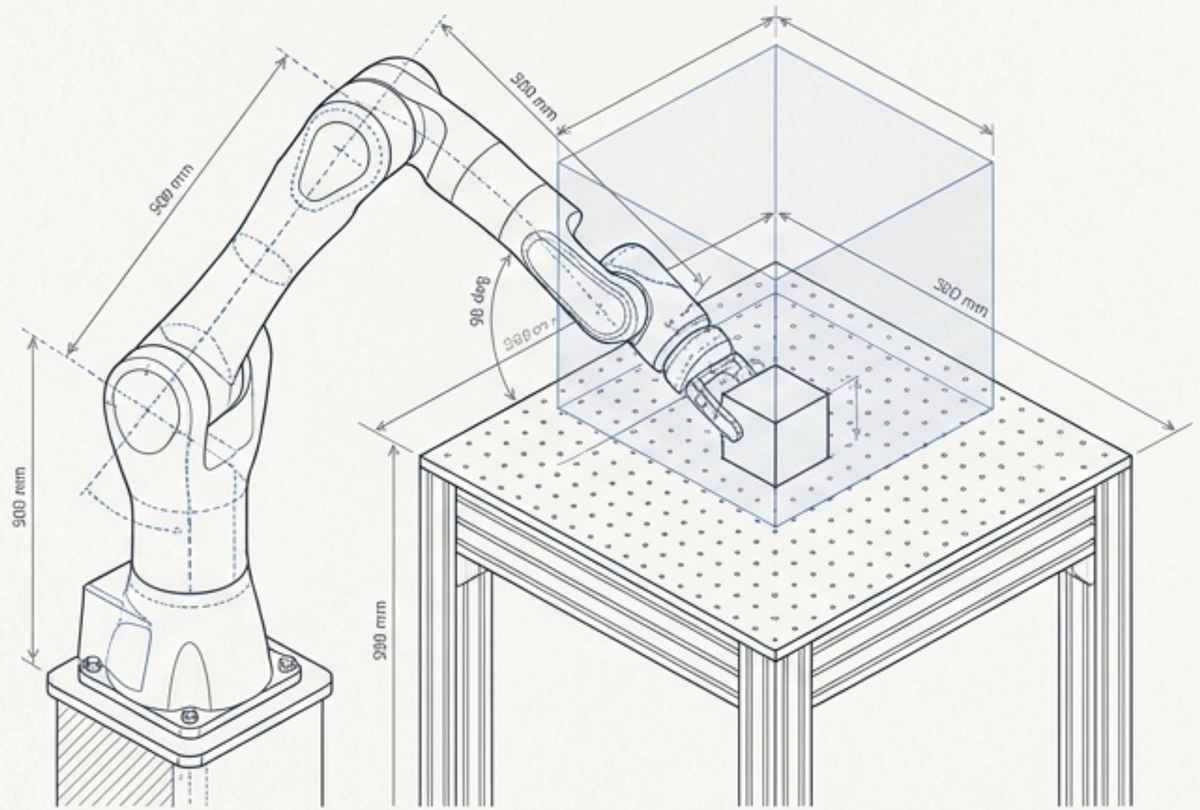
Native integration with Nebius Token Factory APIs enables zero-friction, deterministic data passing.



Gemini Notebook

Tier 1 establishes the deterministic baseline of physical reality using NVIDIA Isaac Lab.

Powered by PhysX 5 / Omniverse to capture exact trajectories, friction coefficients, and SDF bounds.



Data Contract

```
{
  "task": "Isaac-Lift-Cube-Franka-v0",
  "mass": 0.1,
  "friction_coefficient": 0.65,
  "restitution": 0.15,
  "stages": [
    "approach",
    "pre-grasp",
    "grasp",
    "lift",
    "hold"
  ]
}
```

Gemini Notebook

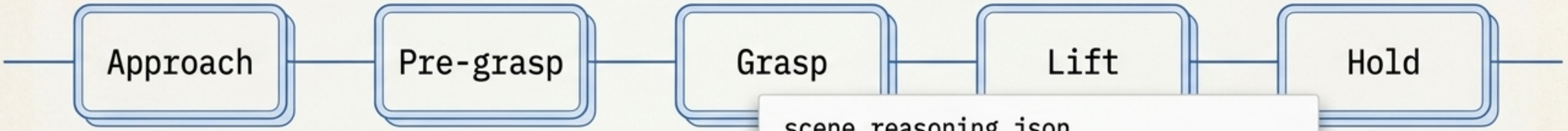
Five bespoke 3D benchmarks isolate specific Newtonian mechanics.

Scenario Task ID	Physical System & Mass	Key Mathematical Constraints
Isaac-Lift-Cube	7-DOF Franka + 0.10 kg Cube	Antipodal force closure: $F_n \geq \frac{m\,g}{2\,\mu}$
Warehouse-AMR	45.0 kg Differential AMR	Floor plane constraint: $z = 0.23$ m
Block-Stack	2.5 kg Cantilevered block	Gravitational torque: $\tau = r \times (m\,g)$
Projectile-3D	0.8 kg Steel Sphere	Restitution energy decay: $E_{k(\text{out})} \leq e^2 \cdot E_{k(\text{in})}$
Ramp-Roll	1.0 kg Rubber Sphere	Inclined acceleration: $a = \frac{5}{7} \cdot g \cdot \sin(\theta)$

Gemini Notebook

Tier 2 utilizes Cosmos 3 to predict multi-stage scene states and manipulation plans.

Ingests **RGB keyframes** to **synthesize trajectories with injected evaluation profiles:** solvent, default hallucination, and severe stress-test.

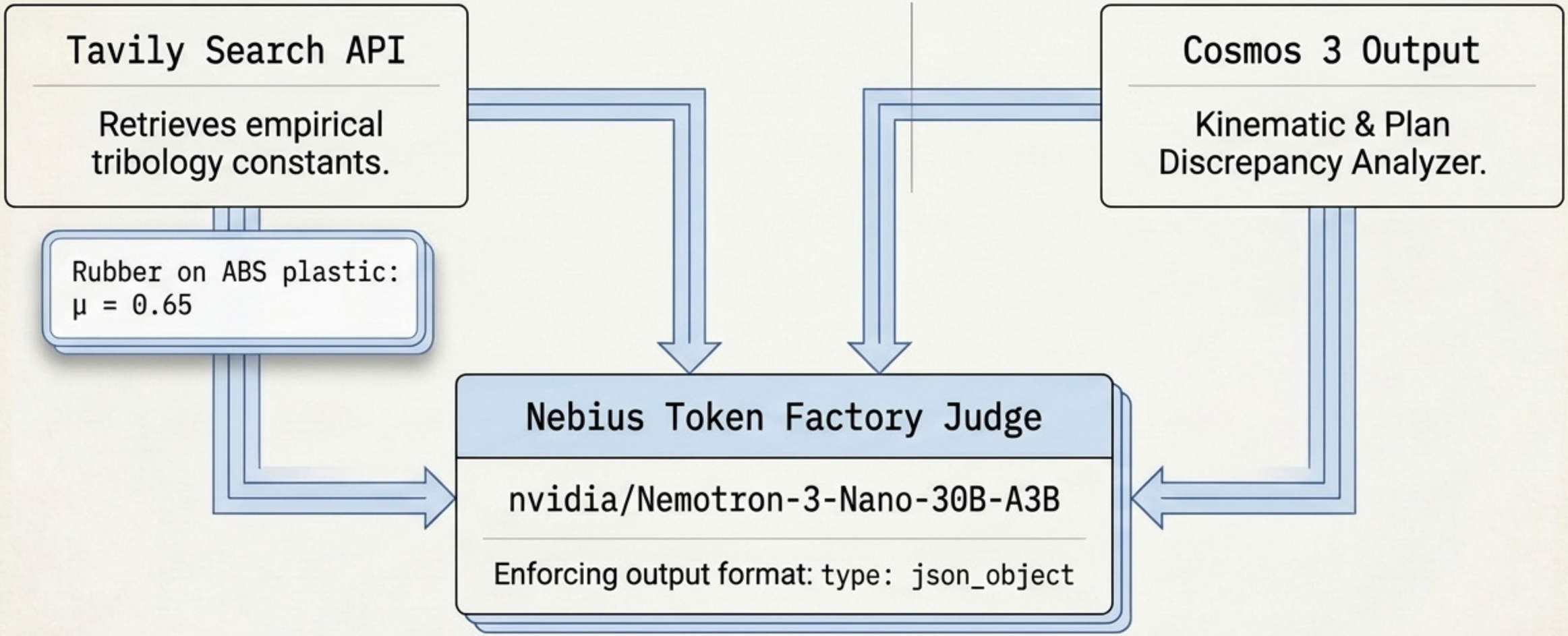


scene_reasoning.json

```
{
  "predicted_ee_z_m": 0.52,
  "predicted_grasp_force_n": 0.0,
  "predicted_sdf_clearance_m": 0.18
}
```

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Tier 3 fuses empirical tribology data with Nemotron reasoning to audit physical laws.



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The automated judge evaluates strict mathematical boundaries across five physical domains.

5 Physical Laws Diagnostic Matrix

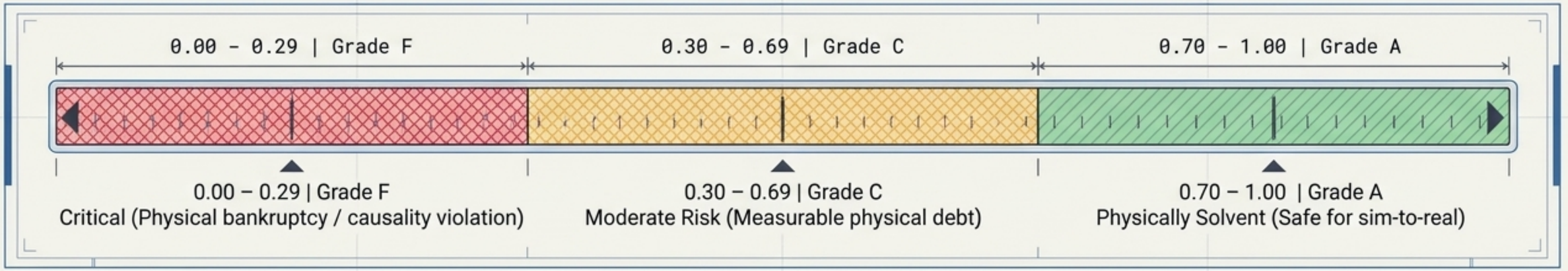
Physical Law	Equation	Violation Condition
Contact Constraint	`SDF(t) >= 0.0 m`	SDF < -0.015 m (Solid-body interpenetration)
Friction Slipping	`F_grasp(t) >= (m*g) / (2*μ)`	Insufficient force closure
Gravity Support	`Velocity_z > 0 ⇒ Support Active`	Ungrounded levitation
Momentum Conservation	`E_pred(t) <= E_GT(t) + ε`	Uncommanded kinetic energy creation
Object Permanence	`visible(t) == True`	Spontaneous bounding-box loss

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The SIPA rubric quantifies physics hallucinations into an actionable integrity rating.

$$PHR = \frac{Violated\ Steps}{Total\ Steps}$$

$$PIR = 0.45(1 - PHR) + 0.25(1 - D_{\text{sev}}) + 0.15(TCS) + 0.15(1 - P_{\text{debt}})$$

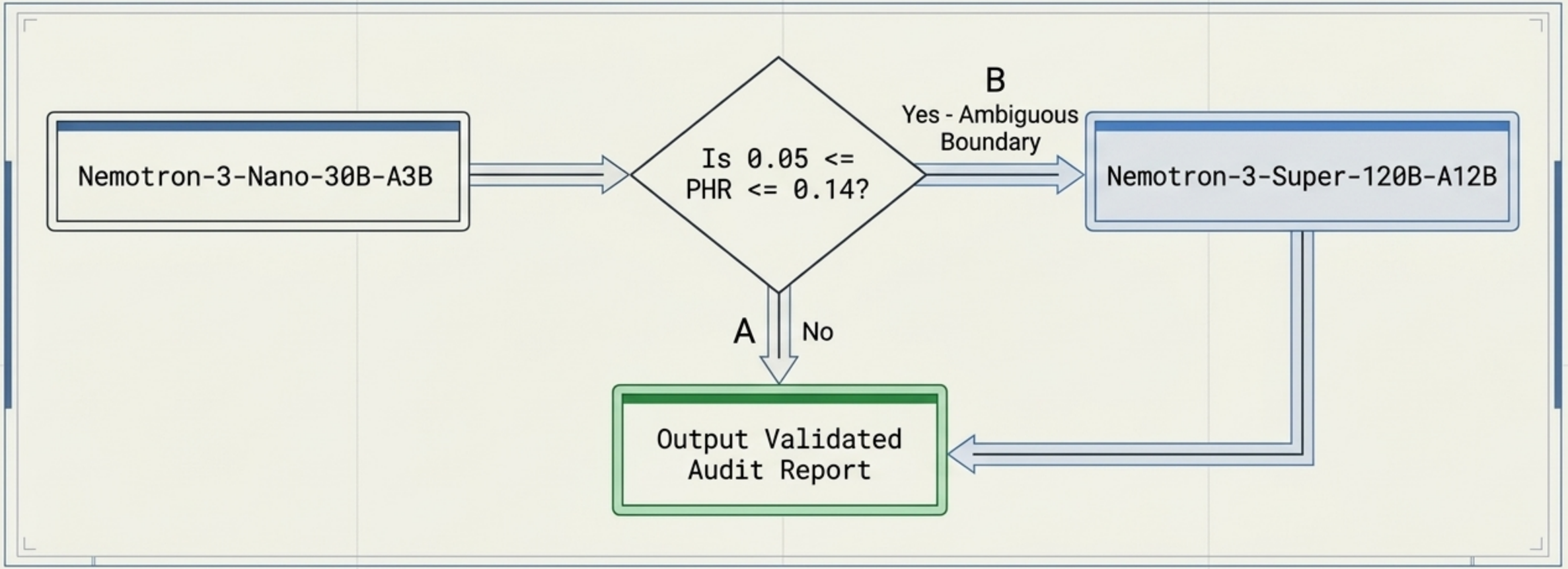


Physical Integrity Rating (PIR) Spectrum

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Ambiguous boundary cases automatically escalate to a Super-Reasoner model.

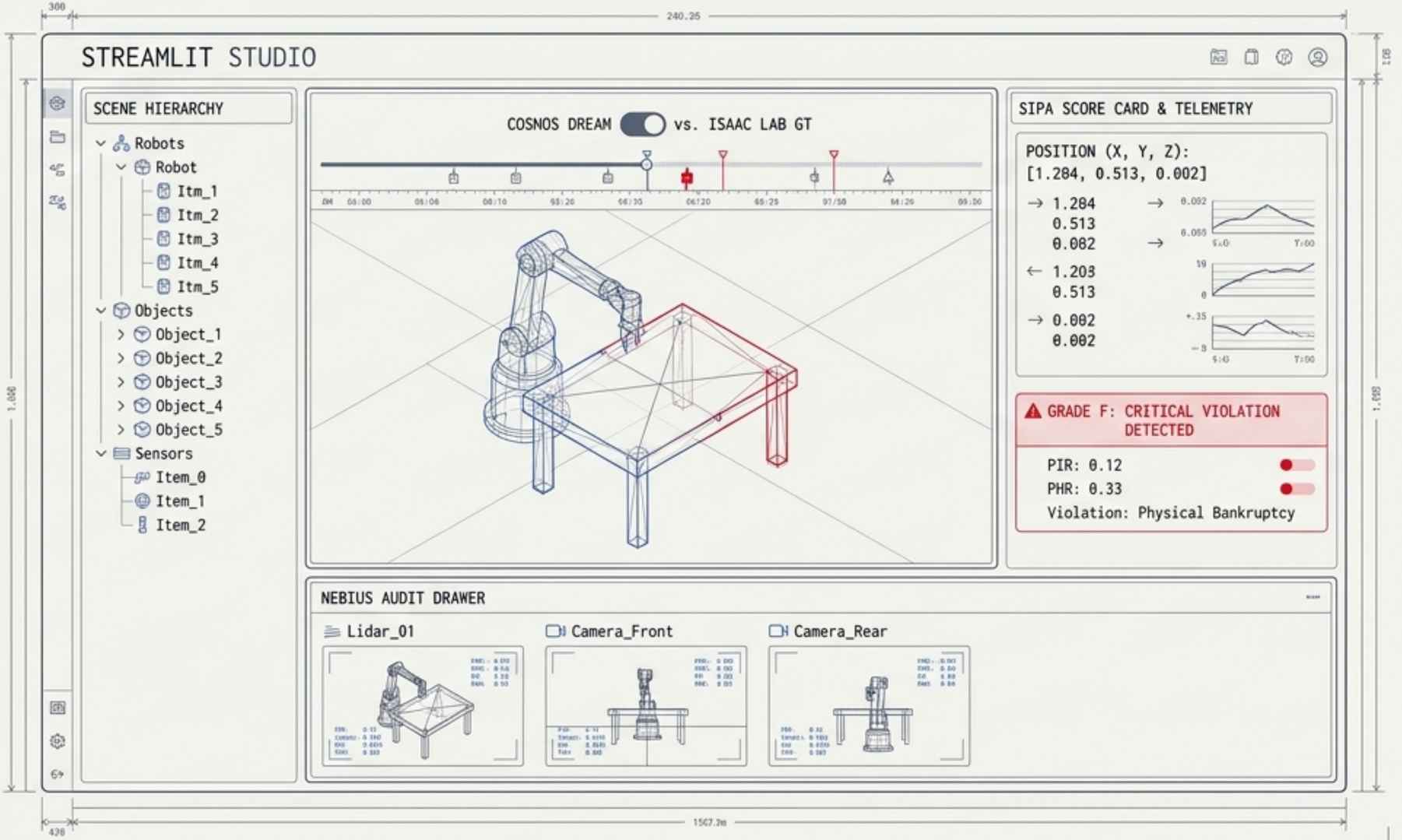
Smart resource routing preserves iteration speed for obvious successes/failures while deploying maximum compute for edge cases.



Gemini Notebook

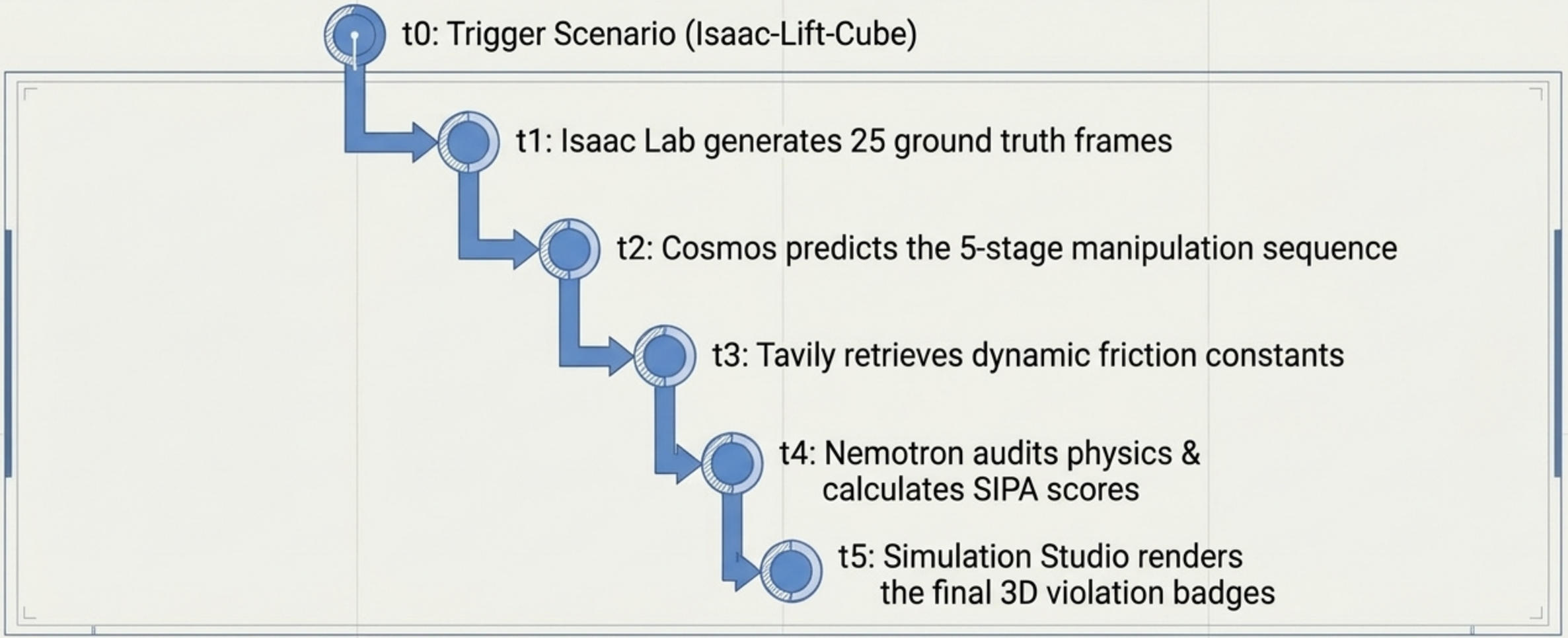
Tier 4 provides an interactive Omniverse viewport to visualize physical violations.

A 3D simulation studio allowing interactive timeline scrubbing and live token-factory auditing.



Gemini Notebook

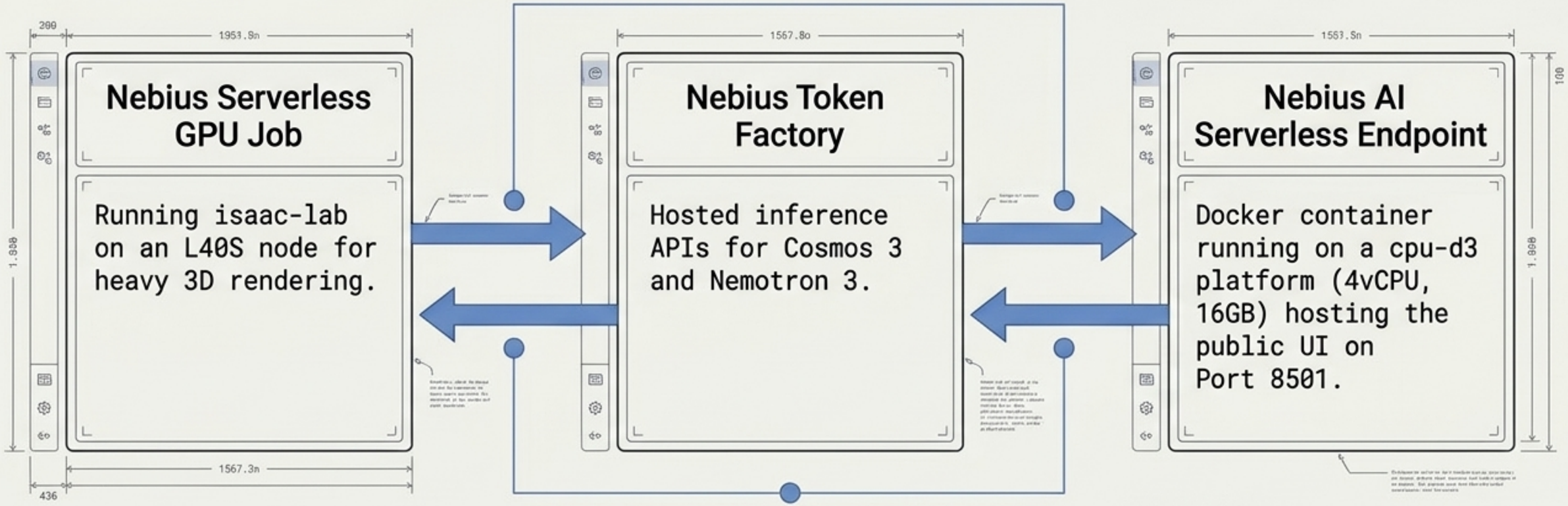
A single API request triggers a chronological, five-step deterministic audit.



Gemini Notebook

Cloud deployment separates GPU-bound rendering from lightweight CPU reasoning.

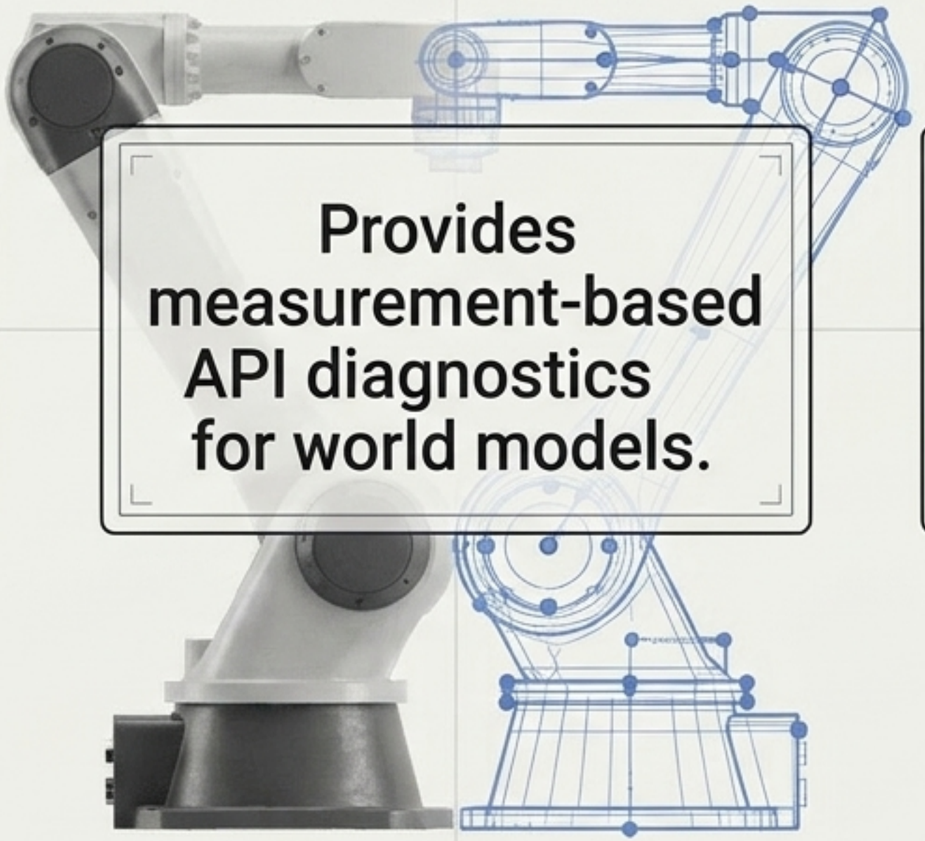
Containerized architecture drastically reduces iteration cost while keeping the interactive dashboard highly responsive.



Gemini Notebook

Deterministic diagnostics for sim-to-real deployment.

Closes the loop between visual perfection and physical reality.



Provides measurement-based API diagnostics for world models.

Pinpoints exactly why a model fails before transferring to hardware.

Built for Nebius Physical AI Hackathon | GitHub: physics-audit

Gemini Notebook